

OpenLDAP Master-Replica TLS Deployment Guide

Prerequisites

- RHEL 9 on both master and replica nodes
- Symas SOLDAP repo accessible via Red Hat Satellite (soldap-release26)
- Root or sudo access on both nodes
- Both nodes can reach each other on port 389 (internal network)
- Master IP: 172.23.11.236, Replica IP: 172.23.11.237 (update as needed)

Files Included

File	Purpose
script/	All master + replica installer scripts
script/0-clean-openldap.sh	Wipes old Symas install for clean start
script/install-symas-openldap-all-in-one.sh	Master orchestrator
script/install-symas-openldap-replica-all-in-one.sh	Replica orchestrator
script/replica/	r1..r9 replica scripts
script/test/	Test scripts

Credentials

Item	Value
Base DN	dc=eab,dc=bank,dc=local
Admin DN	cn=admin,dc=eab,dc=bank,dc=local
Admin Password	TheN1le1
Replicator DN	cn=replicator,dc=eab,dc=bank,dc=local
Replicator Password	replpass

Step-by-Step Deployment

1. Copy scripts to both nodes

```
# From jump station
unzip openldap-tls-deploy.zip -d /tmp/
scp -r /tmp/openldap-tls-deploy/script root@172.23.11.236:/tmp/script
scp -r /tmp/openldap-tls-deploy/script root@172.23.11.237:/tmp/script
```

2. Clean both nodes

```
# On master (172.23.11.236)
ssh root@172.23.11.236 'sudo bash /tmp/script/0-clean-openldap.sh'

# On replica (172.23.11.237)
ssh root@172.23.11.237 'sudo bash /tmp/script/0-clean-openldap.sh'
```

3. Install master with TLS

```
ssh root@172.23.11.236 'sudo TLS_MODE=yes bash /tmp/script/install-symas-openldap-all-in-one.sh'
```

Wait ~2-3 minutes. All tests should pass.

4. Extract master CA cert for replica

```
# Copy CA cert+key from master to jump station
ssh root@172.23.11.236 'sudo cat /opt/symas/etc/openldap/tls/ca.crt' > /tmp/master-ca.crt
ssh root@172.23.11.236 'sudo cat /opt/symas/etc/openldap/tls/ca.key' > /tmp/master-ca.key

# Push to replica
scp /tmp/master-ca.crt /tmp/master-ca.key root@172.23.11.237:/tmp/
```

5. Install replica with TLS (using master CA)

```
MASTER_IP=172.23.11.236

ssh root@172.23.11.237 \
"sudo MASTER_IP=$MASTER_IP ADMIN_PW=TheN1le1 REPL_PW=replpass \
  TLS_MODE=yes COPY_FROM_MASTER=1 \
  STAGED_CA_CERT=/tmp/master-ca.crt \
  STAGED_CA_KEY=/tmp/master-ca.key \
  LDAPTLS_REQCERT=never \
  bash /tmp/script/install-symas-openldap-replica-all-in-one.sh"
```

Wait ~2-3 minutes. You should see PASS=10 FAIL=0 WARN=0 at the end.

6. Test connections

```
# From jump station, extract CA cert for client use:
ssh root@172.23.11.236 'sudo cat /opt/symas/etc/openldap/tls/ca.crt' > /tmp/master-ca.crt
export LDAPTLS_CACERT=/tmp/master-ca.crt

# Master bind
ldapwhoami -x -ZZ -H ldap://172.23.11.236:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1"

# Replica bind
ldapwhoami -x -ZZ -H ldap://172.23.11.237:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1"

# List entries
ldapsearch -x -ZZ -H ldap://172.23.11.236:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1" \
-b "dc=eab,dc=bank,dc=local" -s one "(objectClass=*)" dn
```

7. Test replication

```

export LDAPTLS_CACERT=/tmp/master-ca.crt
TEST_UID="repltest-$(date +%Y%m%d%H%M%S)"

# Add on master
ldapadd -x -ZZ -H ldap://172.23.11.236:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1" <<EOF
dn: uid=${TEST_UID},ou=Users,dc=eab,dc=bank,dc=local
objectClass: inetOrgPerson
objectClass: organizationalPerson
objectClass: person
cn: ReplTest
sn: Test
uid: ${TEST_UID}
userPassword: TestPass123!
EOF

# Wait for replication
sleep 8

# Verify on replica
ldapsearch -x -ZZ -o ldif-wrap=no -H ldap://172.23.11.237:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1" \
-b "uid=${TEST_UID},ou=Users,dc=eab,dc=bank,dc=local" -s base dn

# Cleanup on master
ldapdelete -x -ZZ -H ldap://172.23.11.236:389 \
-D "cn=admin,dc=eab,dc=bank,dc=local" -w "TheN1le1" \
"uid=${TEST_UID},ou=Users,dc=eab,dc=bank,dc=local"

```

CA Certificate for Clients

The master CA cert (/opt/symas/etc/openldap/tls/ca.crt) signs both server certs. Clients must trust it:

```

# Linux/OpenLDAP CLI
export LDAPTLS_CACERT=/path/to/master-ca.crt

# macOS (keychain)
sudo security add-trusted-cert -d -r trustRoot \
-k /Library/Keychains/System.keychain master-ca.crt

# Java/Apache Directory Studio
keytool -importcert -noprompt -trustcacerts -alias ldap-ca \
-file master-ca.crt -keystore truststore.jks -storepass changeit

```

Troubleshooting

Symptom	Fix
slappasswd missing	Packages not installed -- re-run 0-clean + master install
ldap_sasl_interactive_bind_s: Confidentiality req	TLS required -- add -ZZ + CA cert
Replica shows 0 entries	Check syncrepl: ldapsearch -Y EXTERNAL -H ldapi:/// -b cn=config "(olcSyncre
objectClass: value #0 invalid on replica	ppolicy module not loaded -- re-run replica installer
Ldapsearch wraps lines	Add -o ldif-wrap=no
Firewall blocks port 389/636	firewall-cmd --add-port=389/tcp --add-port=636/tcp --permanent && firewall-cm
simple_bind=128 blocks plain LDAP	Use -ZZ (StartTLS) or -H ldaps:// (LDAPS) for password binds

Deployment Summary

Role	IP	Ports	TLS
Master	172.23.11.236	389 (StartTLS), 636 (LDAPS)	Self-signed CA
Replica	172.23.11.237	389 (StartTLS), 636 (LDAPS)	Signed by master CA